

Market-Neutral Trading Algorithm

**Historical graph comparison, statistical uncertainty
and reproducible research**

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Quant Projects

Milestone 3 working paper

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Research question

Can relationships between stock returns improve a constrained reversal strategy after trading costs, borrowing costs and market exposure are accounted for?

This version evaluates the historical graph candidate against the baseline. Persistent homology and the final holdout remain subsequent milestones.

Abstract

This project develops a Python framework for a market-neutral equity trading study. A baseline reversal signal is constructed from recent volatility-scaled returns, and portfolios are constrained to approximately zero dollar exposure and zero exposure to estimated market beta. A weighted correlation graph and Laplacian diffusion provide the proposed extension. The present milestone moves from synthetic verification to an exploratory historical study of 24 surviving US companies, with SPY as the benchmark.

The acquired dataset contains 2,768 aligned sessions from 2 January 2009 to 31 December 2019. The year 2009 supplies estimation history; 2010-2016 is the development period and 2017-2019 is validation. The final period, 2020-2025, has not been acquired or evaluated. At 5 basis points per dollar traded and 2% annual short borrow, the baseline produces annualised net returns of -3.10% in development and -5.83% in validation. The corresponding maximum drawdowns are -24.99% and -18.49%.

The fixed graph candidate returns -4.37% annualised in development and -5.27% in validation. Its validation CAGR improvement over the baseline is 0.55 percentage points, but the 95% stationary-bootstrap interval for the annualised arithmetic daily-return difference spans -1.16 to +2.37 percentage points. Both primary strategies remain unprofitable under the stated costs. Forty-four predeclared backtests cover fees, diffusion time, borrow and matched-gross controls; no new primary model is selected.

Current-vintage adjusted data and a fixed survivor basket limit generalisation. The paper records source-linked code, numbered equations, author-led references and reproducible interactive graph explanations. The next milestone will examine topological descriptors against simpler controls while retaining the reserved final period.

Keywords: statistical arbitrage; market neutrality; graph Laplacian; diffusion; historical backtesting; reproducibility.

Table of contents

Abstract	2
Table of contents	3
List of tables and figures	4
List of equations and code listings	5
List of abbreviations	6
List of mathematical symbols	7
1. Introduction and research context	8
1.1. Questions and scope	8
2. Historical data and experimental protocol	9
2.1. Data audit and sample limitations	10
3. Baseline signal and market sensitivity	11
3.1. Beta estimation and interpretation	12
4. Portfolio construction and exposure limits	13
5. Graph relationships and diffusion	14
5.1. Interactive view of the historical structure	15
6. Execution timing and portfolio accounting	16
7. Historical baseline results	17
7.1. Capital paths and cost sensitivity	18
7.2. Interpretation and remaining uncertainty	19
8. Historical graph comparison	20
8.1. Primary historical outcomes	21
8.2. A causal control for gross exposure	22
8.3. Paired uncertainty with dependent observations	23
8.4. HAC cross-check and interpretation	24
8.5. Sensitivity and the economic effect of costs	25
8.6. Gross contribution, borrow and turnover	26
8.7. Interpreting diffusion through an interactive graph	27
9. Planned persistent-homology extension	28
10. Reproducibility and subsequent milestones	29
References	30

List of tables and figures

Tables

Table 1. Abbreviations used in the working paper	6
Table 2. Mathematical symbols and definitions	7
Table 3. Declared equity universe; research groups are not historical index classifications	9
Table 4. Chronological allocation fixed before baseline performance was inspected	9
Table 5. Data audit for 24 stocks and the SPY benchmark	10
Table 6. Baseline settings retained from the synthetic prototype	11
Table 7. Baseline performance at 5 bps and 2% annual short borrow	17
Table 8. All predeclared trading-cost scenarios; short borrow remains 2%	19
Table 9. Predeclared comparison and sensitivity design	20
Table 10. Primary strategies at 5 bps trading fees and 2% borrow	21
Table 11. Separate backtests of the matched-gross controls	22
Table 12. Pointwise uncertainty for the primary paired-return difference	24
Table 13. Borrow sensitivity: CAGR with trading fees fixed at 5 bps	26
Table 14. Research artifacts and completion state	29

Figures

Figure 1. Historical 126-session correlation matrix at the last validation date	15
Figure 2. Net historical capital and drawdown; phases start with separate capital	18
Figure 3. Sensitivity to trading costs and rolling realised market beta	19
Figure 4. Baseline and graph capital paths under identical accounting conventions	21
Figure 5. Stationary-bootstrap and HAC intervals for the paired mean	25
Figure 6. Graph-minus-baseline CAGR across every declared fee and diffusion time	25
Figure 7. Annualised arithmetic return contributions and recurring charges	26
Figure 8. A historical graph and the resulting unconstrained scores	27

List of equations and code listings

Equations

Equation 1. Simple adjusted-price return	10
Equation 2. Volatility-scaled recent movement	11
Equation 3. Baseline reversal score	11
Equation 4. Trailing market-beta estimate	12
Equation 5. Projection away from dollar and estimated-beta exposure	13
Equation 6. Target portfolio constraints	13
Equation 7. Scaled combinatorial graph Laplacian	14
Equation 8. Graph heat diffusion	14
Equation 9. Proposed graph-relative reversal score	14
Equation 10. Short-borrow accrual	16
Equation 11. Post-cost NAV and drift-aware trading cost	16
Equation 12. Net-return reconciliation	16
Equation 13. Descriptive annualised Sharpe statistic	18
Equation 14. Drawdown from the running capital peak	18
Equation 15. Matching the gross exposure of the two decision portfolios	22
Equation 16. Paired daily difference and annualised arithmetic mean	23
Equation 17. Random restarts and circular continuation in the stationary bootstrap	23
Equation 18. Bartlett-weighted long-run variance of the paired difference	24
Equation 19. HAC standard error of the annualised arithmetic difference	24
Equation 20. Spectral gain applied to the unsmoothed residual	27
Equation 21. Correlation-derived distance for the proposed topology study	28

Code listings

Listing 1. Constructing the historical reversal score	12
Listing 2. Preserving neutrality while enforcing a position ceiling	13
Listing 3. Delaying decisions before slicing the evaluation phase	16
Listing 4. Solving for equity after transaction costs	17
Listing 5. Constructing the graph-relative score from the same observed shock	20
Listing 6. Matching exposure without leveraging either portfolio upward	22
Listing 7. Preserving blocks through random restarts and circular continuation	23
Listing 8. Accumulating serial covariance with Bartlett weights	24

List of abbreviations

Table 1. Abbreviations used in the working paper

Abbreviation	Meaning
ACT/365	Actual calendar days divided by 365
bps	Basis points; one basis point equals 0.01%
CAGR	Compound annual growth rate
CI	Confidence interval
CAPM	Capital Asset Pricing Model
CRSP	Center for Research in Security Prices
CSCV	Combinatorially symmetric cross-validation
CSV	Comma-separated values
ETF	Exchange-traded fund
H0 / H1	Homology dimensions zero and one
HAC	Heteroskedasticity and autocorrelation consistent
NAV	Net asset value
OLS	Ordinary least squares
P&L	Profit and loss
PBO	Probability of backtest overfitting
pp	Percentage points
SB	Stationary bootstrap
PCA	Principal component analysis
SHA-256	Secure Hash Algorithm, 256-bit digest
SPY	Ticker used for the S&P 500 ETF benchmark
TDA	Topological data analysis
USD	US dollars

List of mathematical symbols

Table 2. Mathematical symbols and definitions

Symbol	Definition
$i, j; t; n$	Asset indices; session index; number of assets
$P; r$	Input adjusted price; simple holding-period return
$h; \hat{\sigma}$	Signal horizon; estimated daily return standard deviation
$x; s$	Volatility-scaled shock; unnormalised trading score
$\beta; \hat{\beta}$	Market sensitivity; its trailing estimate
$A; A^+$	Exposure matrix; Moore-Penrose pseudoinverse
$q; w$	Projected score; portfolio weights relative to post-cost NAV
$G; b$	Maximum gross exposure; maximum absolute name weight
$\rho; W; D$	Pairwise correlation; adjacency matrix; degree matrix
$\bar{d}; L; \tau$	Mean weighted degree; scaled Laplacian; diffusion time
$\tilde{x}; d(i,j)$	Diffused signal; correlation-derived distance
$v; V; c$	Signed asset notional; NAV; cost per dollar traded
$a; \Delta$	Annual borrow rate; elapsed calendar days
$B; C$	Borrow cost; transaction cost
$R; S; DD$	Portfolio return; descriptive Sharpe statistic; drawdown
$G^*; B; G$	Common gross exposure; baseline/graph superscript labels
$\delta; \mu; T$	Paired daily difference; annualised arithmetic mean; phase sample size
$\ell; l^*$	Expected bootstrap block length; resampled observation index
$K; \gamma; \Omega$	HAC lag count; lagged covariance; long-run variance
$SE; \lambda$	Standard error; graph-Laplacian eigenvalue

1. Introduction and research context

As Avellaneda and Lee explain in *Statistical arbitrage in the US equities market* (2010), systematic equity strategies can use factor-adjusted residual behaviour to form market-neutral portfolios [1]. Their PCA- and ETF-based construction provides context for this research. The present baseline and graph residual are different models, and no result from their paper is transferred to this stock basket.

Lo and MacKinlay, in *When Are Contrarian Profits Due to Stock Market Overreaction?* (1990), show that contrarian profits need not arise solely from overreaction [2]. This distinction is relevant because reversing a recent stock movement is a trading hypothesis, not proof that its price is incorrect. Accordingly, the first historical experiment retains a simple baseline and evaluates its costs and exposures before adding graph complexity.

The project builds on the earlier SPX density study in its use of controlled numerical checks, reusable Python modules and explanatory notebooks. Its empirical objective is different: it evaluates a trading rule rather than recovering an option-implied probability density. The final research presentation will include a reproducible GitHub repository and an interactive companion page suitable for sharing through LinkedIn.

1.1. Questions and scope

Three questions guide the work. First, does the baseline remain viable after explicit costs? Second, does graph diffusion add information under an otherwise identical experiment? Third, can persistent-homology features add useful regime information beyond simpler measures such as volatility and mean correlation? Milestones 2 and 3 evaluate the first two questions. Persistent homology remains a planned extension.

The experimental code and the scientific references have separate roles. The references justify established concepts and numerical relationships; parameter settings, the fixed basket and the proposed graph score are choices made for this project. The historical comparison below assesses the fixed graph candidate without claiming that its parameter values are justified by the references.

2. Historical data and experimental protocol

Table 3. Declared equity universe; research groups are not historical index classifications

Research group	Stock tickers
Technology	AAPL, MSFT, IBM, ORCL
Financials	JPM, BAC, WFC, GS
Energy	XOM, CVX, COP, SLB
Healthcare	JNJ, PFE, MRK, UNH
Consumer retail	WMT, COST, TGT, HD
Consumer brands	KO, PEP, MCD, SBUX

The primary observations are Yahoo Finance daily chart responses [15]. The analysis uses the supplied **adjclose** field; close, volume, dividend events and split events are retained for audit. The data are current-vintage vendor observations rather than a point-in-time archive. Historical adjustment accuracy has not been independently certified, and dividends are not added a second time to adjusted-price returns.

Table 4. Chronological allocation fixed before baseline performance was inspected

Period	Dates	Use
Warm-up	2009	Trailing estimates only
Development	2010-2016	Baseline construction and diagnostic evaluation
Validation	2017-2019	Declared chronological check; not a final test
Final holdout	2020-2025	Reserved; no bars acquired or strategy results inspected in this milestone

As White explains in *A Reality Check for Data Snooping* (2000), repeated model selection on the same observations can create apparently favourable findings by chance [9]. Therefore, the universe, dates, cost scenarios and baseline settings were written to a local protocol and hashed before the equity study was run. This record is not external preregistration, and it does not eliminate knowledge of historical market events.

2.1. Data audit and sample limitations

Table 5. Data audit for 24 stocks and the SPY benchmark

Audit item	Observed result
Sessions per series	2768
First / last session	2 January 2009 / 31 December 2019
Exact calendar agreement	All 25 series match SPY observations
Missing values	0
Nonpositive volume	0
Absolute adjusted returns above 40%	0
Longest unchanged-price run	2 sessions
Dividend / split events	1081 / 4 retained provider events
Final holdout	Absent from the permitted historical cache

Shumway, in *The Delisting Bias in CRSP Data* (1997), documents the importance of omitted adverse delisting returns [5]. This project has a broader selection limitation: its named basket contains companies that survived to the selection date. It is consequently an exploratory fixed-basket study and cannot establish performance for the historical investable equity universe.

The loader does not silently forward-fill missing prices or intersect mismatched calendars. Positive prices and reported volumes are required throughout. A 20-session close-times-volume proxy is reported, but this is not a calibrated capacity limit. Borrow availability, suspended securities and a complete corporate-action reconstruction remain unresolved for a production equity simulator.

$$r_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1 \quad \text{Equation 1}$$

Equation 1 defines the input return. Where: P is the supplied adjusted-price series, i denotes the asset and t denotes the observed session. Avellaneda and Lee also formulate their historical return analysis using dividend-adjusted prices [1]. Here the resulting return-bearing notionals are an approximation; the ledger does not model raw execution prices and dividend cash flows separately.

3. Baseline signal and market sensitivity

Table 6. Baseline settings retained from the synthetic prototype

Setting	Value
Recent-movement horizon	5 sessions
Volatility estimate	63 sessions; sample standard deviation
Market beta / correlation window	126 sessions
Shock clipping	-5 to +5 scaling units
Gross / name exposure ceilings	100% / 8% of post-cost NAV
Base trading cost	5 bps per dollar bought or sold
Alternative trading costs	0, 10 and 20 bps
Annual short borrow	2%, ACT/365
Initial capital per phase	USD 100,000
Execution	Decision at t; trade at t+1 close; first new-position P&L at t+2

$$x_{i,t} = \text{clip} \left(\frac{\sum_{u=t-h+1}^t \log(1 + r_{i,u})}{\hat{\sigma}_{i,t} \sqrt{h}}, -5, 5 \right) \quad \text{Equation 2}$$

Equation 2 scales recent movement by trailing volatility. Where: $h = 5$ and $\sigma\text{-hat}$ is the 63-session daily sample standard deviation. This is a project scaling rule, not a calibrated normal-distribution z-score. The five-day horizon and clipping bounds were not inferred from the cited papers or optimised on these historical results.

$$s_{i,t}^{\text{base}} = -x_{i,t} \quad \text{Equation 3}$$

Equation 3 assigns a negative raw score to a positive recent movement. The sign expresses the reversal hypothesis discussed in Section 1; it is followed by portfolio construction rather than interpreted directly as an order.

Listing 1. Constructing the historical reversal score

```

volatility = r[t + 1 - config.volatility_window:t + 1].std(axis=0, ddof=1)
if (volatility <= 1e-12).any():
    raise ValueError("A constant stock return series needs an explicit eligibility rule")
shock = np.log1p(r[t + 1 - config.signal_window:t + 1]).sum(axis=0)
shock = np.clip(shock / (volatility * np.sqrt(config.signal_window)), -5, 5)
weights.iloc[t] = neutral_weights(-shock, beta, config.gross_limit, config.position_limit)

```

Source: **market_neutral/historical.py**, function **baseline_decisions**. Selected executable lines; surrounding input checks remain in the module.

3.1. Beta estimation and interpretation

In *Capital Asset Prices* (1964), Sharpe develops the relationship between asset risk and market equilibrium [3]. This supplies the context for measuring systematic market sensitivity. The implementation below uses a trailing OLS slope to SPY as an operational estimate; it does not assume that the CAPM describes every stock return.

$$\hat{\beta}_{i,t} = \frac{\sum_{u=t-m+1}^t (r_{i,u} - \bar{r}_i)(r_{M,u} - \bar{r}_M)}{\sum_{u=t-m+1}^t (r_{M,u} - \bar{r}_M)^2} \quad \text{Equation 4}$$

Equation 4 is the intercept-inclusive OLS slope. Where: $m = 126$, M identifies SPY and the bars denote means within that trailing window. A benchmark window with negligible variance is rejected. Each estimate is available at its decision close and is shifted before execution.

Fama and French, in *Common risk factors in the returns on stocks and bonds* (1993), identify common equity variation beyond the overall market factor [4]. Therefore, neutralising a single SPY beta does not establish neutrality to every economic risk. This milestone does not constrain sector, size, value or momentum exposure.

4. Portfolio construction and exposure limits

Golub and Van Loan discuss least-squares and rank-deficient problems in *Matrix Computations* (2013), particularly Chapter 5 [14]. Using that standard projection framework, this project removes the components of each score that load on the constant vector and the estimated market-beta vector. A least-squares solve is used in the code so that identical beta estimates do not require an invertible normal-equations matrix.

$$A_t = [\mathbf{1}, \hat{\beta}_t], \quad q_t = s_t - A_t A_t^+ s_t \quad \text{Equation 5}$$

Equation 5 defines the projected score q . Where: A contains the two exposure directions and A^+ denotes the Moore-Penrose pseudoinverse. The final weights are obtained by uniform rescaling. An effectively zero projected score gives a flat portfolio.

$$\mathbf{1}^T w_t = 0, \quad \hat{\beta}_t^T w_t = 0, \quad \|w_t\|_1 \leq G, \quad \|w_t\|_\infty \leq b \quad \text{Equation 6}$$

Equation 6 states the operational definition of neutrality and the exposure ceilings. Where: $G = 1$ and $b = 0.08$. At full utilisation, gross exposure is approximately 50% long plus 50% short. If the name limit binds, all positions shrink by the same factor; clipping names independently would generally disturb the exposure constraints.

Listing 2. Preserving neutrality while enforcing a position ceiling

```
exposures = np.column_stack([np.ones(len(score)), beta])
neutral = score - exposures @ np.linalg.lstsq(exposures, score, rcond=None)[0]
gross = np.abs(neutral).sum()
if gross <= 1e-10 * max(1.0, np.linalg.norm(score)):
    return np.zeros_like(score)
weights = neutral * (gross_limit / gross)
weights *= min(1.0, position_limit / np.abs(weights).max())
```

Source: **market_neutral/portfolio.py**, function **neutral_weights**. Selected executable lines; surrounding input checks remain in the module.

The constraints apply to post-cost target weights at the rebalance. The beta vector is an estimate from the previous decision close. Subsequent price drift and estimation error can produce realised beta even when the recorded estimated exposure is numerically zero.

5. Graph relationships and diffusion

As Mantegna explains in *Hierarchical structure in financial markets* (1999), a stock-return correlation matrix can support a graph representation that reveals an economically meaningful organisation [6]. His minimum-spanning-tree construction is not reproduced here. The project uses a symmetric union of positive-correlation neighbour selections, preserving up to five directed neighbours above 0.20 before symmetrisation.

Chung introduces graph Laplacians and their spectra in *Spectral Graph Theory* (1997) [7]. The distinction between Laplacian conventions matters: Equation 7 uses the combinatorial $D - W$ matrix divided by the mean weighted degree, rather than the symmetric degree-normalised Laplacian emphasised in much of Chung's treatment.

$$D_{ij} = \sum_j W_{ij}, \quad L = \frac{D - W}{\bar{d}} \quad \text{Equation 7}$$

Equation 7 defines W as the symmetric adjacency matrix and $\bar{d}$ as the mean weighted degree. An empty graph uses $L = 0$. The scale choice is specific to this project and gives the diffusion-time parameter a more comparable magnitude as connectivity changes.

Kondor and Lafferty, in *Diffusion Kernels on Graphs and Other Discrete Input Spaces* (2002), construct graph kernels through matrix exponentiation and connect them to heat diffusion [8]. This motivates the smoothing operator in Equation 8. It does not establish that its residual is a profitable financial signal.

$$\tilde{x}_t = \exp(-\tau L_t) x_t \quad \text{Equation 8}$$

Equation 8 smooths the observed shock over the graph. Where: $\tau = 1$ in the current prototype. The implementation uses a symmetric eigendecomposition and has been checked against a direct matrix exponential, constant preservation and decreasing graph energy.

$$s_t^{\text{graph}} = -(x_t - \tilde{x}_t) \quad \text{Equation 9}$$

Equation 9 is the proposed trading rule. It reverses deviations from the diffused value before applying the same portfolio constraints. The graph score was first checked on synthetic data; its historical comparison is reported in Section 8.

5.1. Interactive view of the historical structure

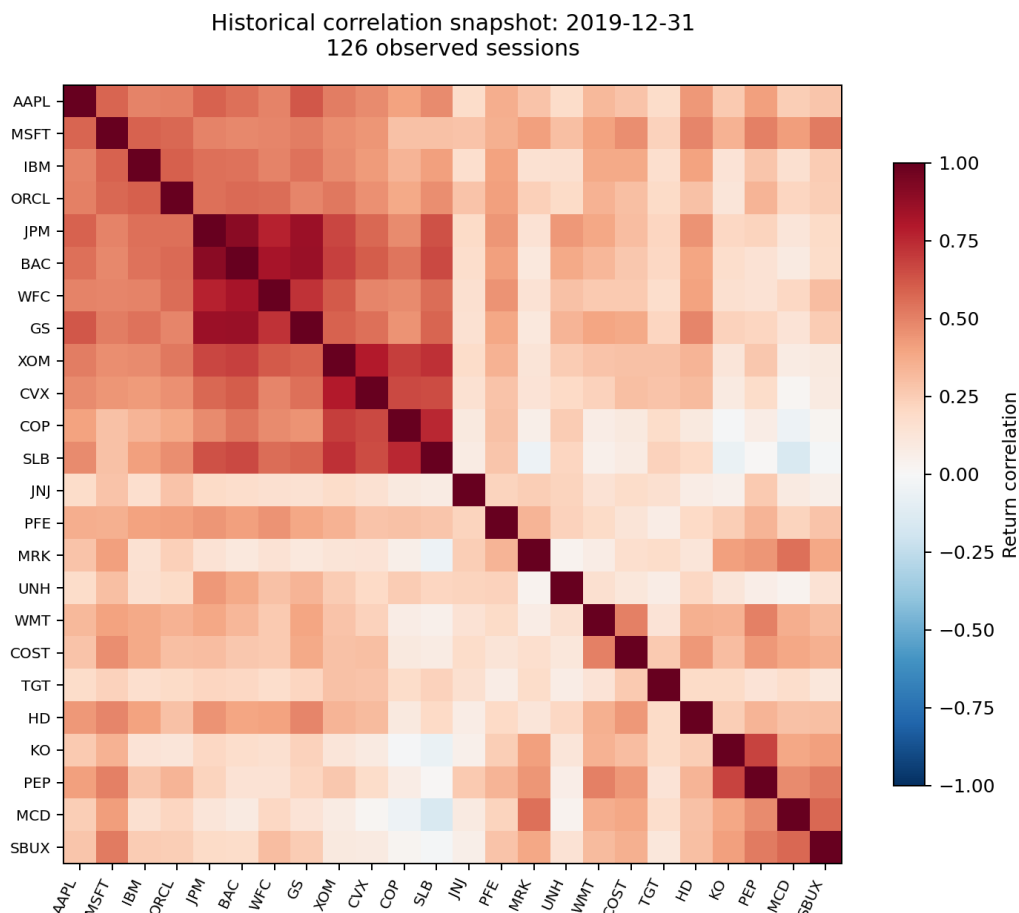


Figure 1. Historical 126-session correlation matrix at the last validation date

Figure 1 gives a static view of the last validation snapshot. The companion **Historical-Correlation-Explorer.html** contains 120 monthly frames, with rotation, zoom, a date slider and playback. Each frame uses trailing observations through the displayed date; no final-holdout data enter the surface.

The surface height is correlation, not expected return. Asset order remains fixed and the asset axes are discrete. The mesh joins neighbouring cells only for display, so values between stock labels should not be interpreted as estimated relationships between intermediate assets. The full correlation matrix is also distinct from the thresholded neighbour graph used by the strategy.

6. Execution timing and portfolio accounting

The accounting convention is deliberately explicit: a close- t observation produces a decision, the decision trades at close $t+1$, and the newly formed position first earns the return ending at close $t+2$. This is a modelling convention chosen to avoid obtaining a close price and simultaneously earning the return ending at that same close. It remains an approximation to executable orders.

Listing 3. Delaying decisions before slicing the evaluation phase

```
execution = weights.shift(1, fill_value=0.)
exposure = betas.shift(1)
mask = (returns.index >= pd.Timestamp(start)) & (returns.index <= pd.Timestamp(end))
```

Source: **market_neutral/historical.py**, function **phase_inputs**. Selected executable lines; surrounding input checks remain in the module.

Listing 3 shows why the shift occurs before period slicing. The first execution in a phase may use the preceding session's valid decision, but each phase begins with a flat book and fresh capital. The final close liquidates positions. Thus neither entry costs nor terminal closing costs disappear at an evaluation boundary.

$$B_t = a \frac{\Delta_t}{365} \sum_i \max(-v_{i,t-1}, 0) \quad \text{Equation 10}$$

Equation 10 charges the annual rate a on the prior close's absolute short notional v . Delta counts elapsed calendar days, including weekends. This uniform borrow assumption is illustrative; it is not a historical security-level borrow series.

$$V_t^{\text{after}} + c \sum_i |w_{i,t} V_t^{\text{after}} - v_{i,t}^{\text{marked}}| = V_t^{\text{before}} \quad \text{Equation 11}$$

Equation 11 solves for NAV after trading fees. Where: c is the per-dollar fee and marked notionals incorporate the day's price changes. NAV before trading already reflects P&L and borrow. The scalar solve makes target weights refer to equity after fees, avoiding a small exposure overshoot caused solely by transaction costs.

$$R_t^{\text{net}} = \frac{PnL_t - C_t - B_t}{V_{t-1}^{\text{after}}} \quad \text{Equation 12}$$

Equation 12 reconciles gross position P&L, trading cost C and borrow cost B to net return. Turnover counts every dollar bought and every dollar sold, divided by start-of-day NAV. Cash interest, financing rebates, margin constraints and nonlinear market impact are not included.

Listing 4. Solving for equity after transaction costs

```
def balance(after):
    return after + cost_rate * np.abs(w * after - marked).sum() - nav_before_trade
if balance(0) >= 0:
    raise ValueError("Cannot fund trading costs")
nav = float(brentq(balance, 0, nav_before_trade, xtol=1e-10)) if cost_rate else nav_before_trade
```

Source: **market_neutral/backtest.py**, function **execute_targets**. Selected executable lines; surrounding input checks remain in the module.

7. Historical baseline results

Table 7. Baseline performance at 5 bps and 2% annual short borrow

Metric	Development	Validation
Total net return	-19.79%	-16.44%
Annualised net return	-3.10%	-5.83%
Annualised volatility	3.17%	3.35%
Zero-cash-rate Sharpe	-0.98	-1.78
Maximum drawdown	-24.99%	-18.49%
Realised SPY beta	0.0202	0.0169
Mean daily turnover	41.00%	39.62%
Mean gross exposure	62.78%	61.29%
Maximum absolute dollar exposure	2.41e-15	3.52e-15
Maximum absolute estimated beta	2.29e-15	3.39e-15

Table 7 reports a negative annualised return in both phases: -3.10% in development and -5.83% in validation. The numerical exposure constraints are satisfied, but these constraints do not imply a viable strategy. The practical issue is whether the signal can earn enough before costs to support its turnover.

Lo, in *The Statistics of Sharpe Ratios* (2002), explains why return dependence and estimation error affect Sharpe-ratio interpretation [11]. The statistic below uses the conventional square-root-of-time scaling and a zero cash rate for descriptive comparison. It is not a dependence-adjusted significance test or a confidence interval.

$$\widehat{S} = \sqrt{252} \frac{\overline{R^{\text{net}}}}{s(R^{\text{net}})}$$

Equation 13

Equation 13 uses the sample daily standard deviation. It is labelled as a zero-cash-rate statistic because cash interest is not modelled. Comparisons with an external fund's published Sharpe would require matching return and financing conventions.

$$DD_t = \frac{V_t}{\max(V_0, V_1, \dots, V_t)} - 1$$

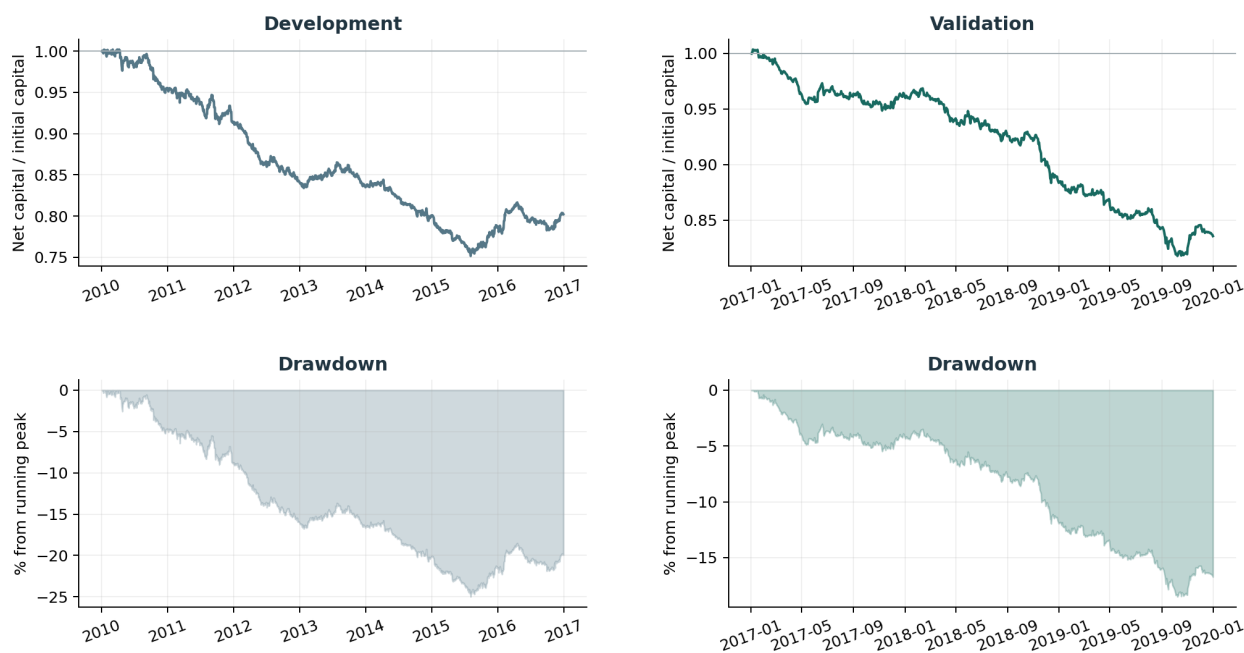
Equation 14

Equation 14 includes initial capital in the running peak, so an early loss is counted. The reported maximum drawdown is the most negative observed value, including the final liquidation date.

7.1. Capital paths and cost sensitivity

Historical baseline: development and validation

Fixed survivor basket · 24 US stocks · 5 bps per dollar traded · 2% annual short borrow



Each phase starts flat and includes entry and final liquidation. Final holdout (2020-2025) remains unacquired and unevaluated.

Figure 2. Net historical capital and drawdown; phases start with separate capital

Table 8. All predeclared trading-cost scenarios; short borrow remains 2%

Cost per dollar	Development CAGR	Validation CAGR
0 bps	2.03%	-1.00%
5 bps	-3.10%	-5.83%
10 bps	-7.98%	-10.41%
20 bps	-17.02%	-18.93%

The zero-trading-fee scenario is not a costless portfolio: it still pays short borrow. Development is positive under that scenario, whereas validation is negative. Consequently, the validation failure cannot be attributed only to the 5 bps trading-fee assumption. The signal itself requires a stronger empirical comparison.

7.2. Interpretation and remaining uncertainty

Trading costs and residual market exposure

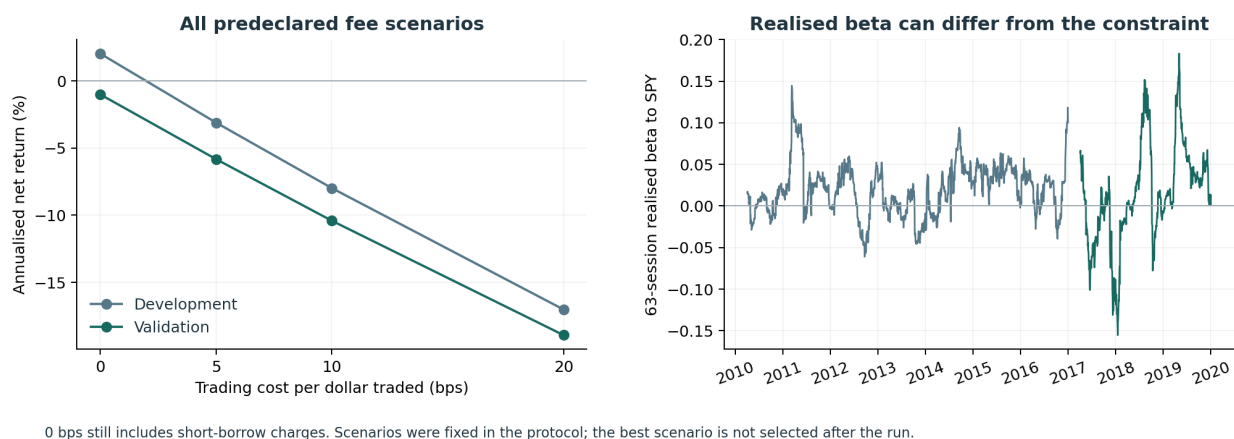


Figure 3. Sensitivity to trading costs and rolling realised market beta

Figure 3 demonstrates two separate issues. Net returns decline as the stated cost per dollar increases. Realised beta also varies through time despite the target constraint. The variation reflects an estimated hedge, a finite rolling measurement window and changing returns; it is not a contradiction of the recorded numerical neutrality checks.

Bailey and colleagues, in *The probability of backtest overfitting* (2017), emphasise that a held-out sample alone does not account for repeated strategy searches [10]. For this reason, the project records each experiment and does not interpret its chronological split as proof against overfitting. No PBO estimate, CSCV analysis or White reality-check statistic has been calculated in this milestone.

The comparison in Section 8 retains these dates, costs, exposure ceilings and execution rules, including the unfavourable results. Any subsequent tuning must be confined to the development process and documented before the reserved final test is released.

8. Historical graph comparison

Milestone 3 evaluates the graph rule in Equation 9 using exactly the acquired observations and accounting conventions from the historical baseline. The baseline outcome was known when this comparison began. The graph settings were inherited from the synthetic prototype, and a separate local protocol was hashed before the first historical graph run. Neither development nor validation is represented as a newly untouched dataset.

Table 9. Predeclared comparison and sensitivity design

Component	Fixed design
Primary candidate	126-session graph; five directed neighbours above 0.20; symmetric union; diffusion time one
Trading-fee sensitivity	0, 5, 10 and 20 bps; 2% annual borrow
Diffusion-time sensitivity	0.5, 1 and 2; every trading-fee combination retained
Borrow sensitivity	0%, 2% and 5%; baseline and primary graph at 5 bps
Exposure control	Both primary targets scaled down to their common daily gross exposure; 5 bps and 2% borrow
Uncertainty	Paired stationary bootstrap, 5,000 replicates; mean block lengths 5/10/20; HAC with 10 lags
Experiment record	44 distinct backtests completed; no replacement of the primary candidate
Final holdout	2020-2025 remains unacquired and unavailable

As White argues in *A Reality Check for Data Snooping* (2000), searching across alternatives can produce misleading apparent success [9]. For this reason, all declared sensitivity results are retained and the primary candidate remains $\tau = 1$. This protocol does not implement White's multiple-comparison test or retrospectively remove earlier knowledge of the baseline.

Listing 5. Constructing the graph-relative score from the same observed shock

```
corr, adjacency, laplacian = correlation_graph(
    r[t + 1 - config.correlation_window:t + 1],
    config.neighbours, config.minimum_correlation)
smooth = heat_diffusion(shock, laplacian, config.diffusion_time)
scores = {"baseline": -shock, "graph": -(shock - smooth)}
```

Source: **market_neutral/strategy.py**, function **build_decisions**. Selected executable lines; surrounding input checks remain in the module.

8.1. Primary historical outcomes

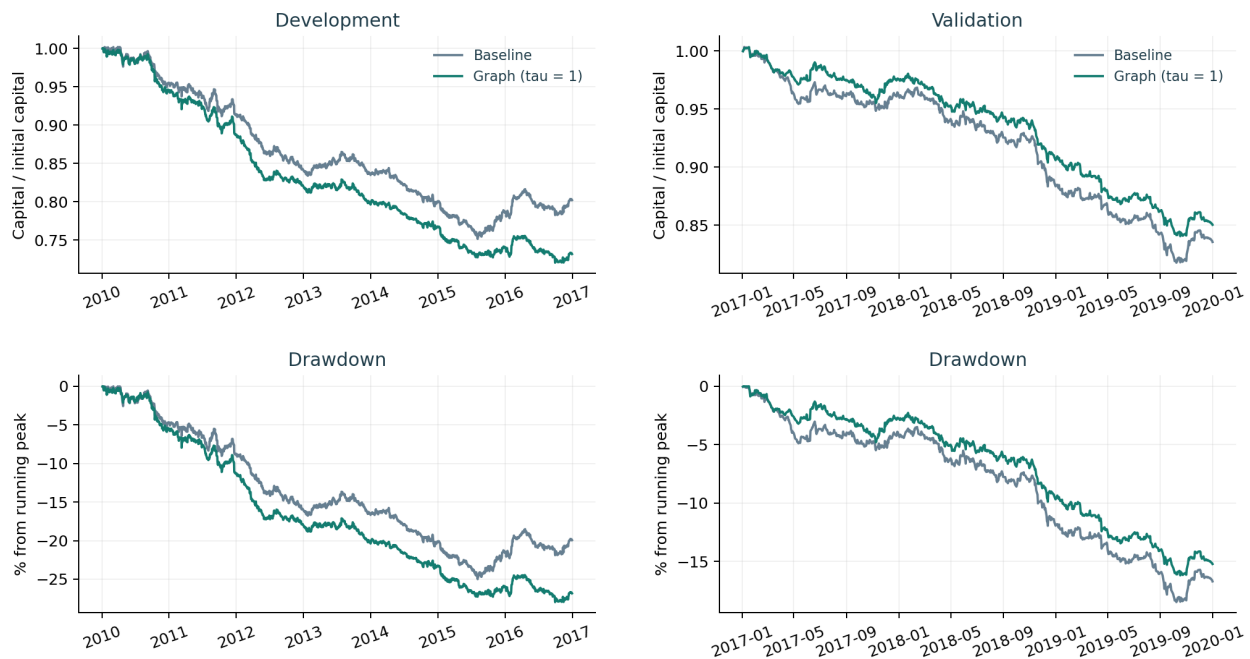
Table 10. Primary strategies at 5 bps trading fees and 2% borrow

Phase / model	Ann. net return	Ann. volatility	Max. drawdown	Mean gross	Mean turnover
Development / Baseline	-3.10%	3.17%	-24.99%	62.78%	41.00%
Development / Graph, tau = 1	-4.37%	2.89%	-27.97%	62.18%	40.44%
Validation / Baseline	-5.83%	3.35%	-18.49%	61.29%	39.62%
Validation / Graph, tau = 1	-5.27%	2.92%	-16.20%	60.63%	39.73%

The graph candidate earns -4.37% annualised in development, compared with -3.10% for the baseline. In validation, the corresponding values are -5.27% and -5.83%. The graph-minus-baseline CAGR differences are therefore -1.26 and +0.55 percentage points. Both strategies lose money at the default costs.

Graph diffusion against the historical baseline

5 bps per dollar traded | 2% annual short borrow | separate capital at each phase boundary



Observed development and validation only. Fixed survivor basket; no final-holdout observations or model selection.

Figure 4. Baseline and graph capital paths under identical accounting conventions

The graph portfolio has somewhat lower realised volatility and lower mean gross exposure. Equal exposure ceilings do not force equal realised capital utilisation. Therefore, the return comparison alone cannot isolate the effect of the graph signal from the portfolio scaling that follows it.

8.2. A causal control for gross exposure

The control below matches the two portfolios at the decision close. It reduces both targets to the smaller of their gross exposures, then applies the original execution delay. It does not scale a portfolio upward, use realised future volatility or relax a name limit. This is a project-specific diagnostic, not a claim that equal gross exposure implies equal risk.

$$G_t^* = \min(\|w_t^B\|_1, \|w_t^G\|_1), \quad w_t^{j,*} = w_t^j \frac{G_t^*}{\|w_t^j\|_1} \quad \text{Equation 15}$$

Equation 15 defines the control for j equal to baseline B or graph G . A zero denominator gives a flat vector. Uniform shrinking preserves dollar and estimated-beta neutrality and cannot increase any absolute name weight. Risk and sector composition can still differ.

Listing 6. Matching exposure without leveraging either portfolio upward

```
gross_baseline = baseline.abs().sum(axis=1)
gross_graph = graph.abs().sum(axis=1)
common = np.minimum(gross_baseline, gross_graph)
controls = []
for weights, gross in [(baseline, gross_baseline), (graph, gross_graph)]:
    factor = common.div(gross.where(gross > 1e-14)).fillna(0.)
    controls.append(weights.mul(factor, axis=0))
```

Source: **market_neutral/comparison.py**, function **match_gross_exposure**. Selected executable lines; surrounding input checks remain in the module.

Table 11. Separate backtests of the matched-gross controls

Phase	Baseline CAGR	Graph CAGR	Difference (pp)	Common mean gross
Development	-3.03%	-4.05%	-1.03	57.65%
Validation	-5.16%	-4.95%	+0.22	56.38%

The validation advantage becomes smaller when gross exposure is matched, while the development disadvantage remains. This reinforces the need to discuss exposure and volatility alongside return. The controls are fully rerun portfolios, not an after-the-fact division of the original returns by average exposure.

8.3. Paired uncertainty with dependent observations

The uncertainty calculation concerns the paired daily net-return difference at the primary cost settings. Both strategies experience the same dates and market observations. Resampling their paired difference preserves that contemporaneous comparison; resampling the two strategies independently would discard it.

$$\delta_t = R_t^{G, \text{net}} - R_t^{B, \text{net}}, \quad \hat{\mu}_{\text{ann}} = 252 \bar{\delta} \quad \text{Equation 16}$$

Equation 16 defines the estimated annualised arithmetic difference. It is not the difference between the two compounded annual growth rates. The uncertainty intervals below refer to this mean, so their estimate should not be substituted for the CAGR difference in the performance table.

Politis and Romano introduce the stationary bootstrap in *The Stationary Bootstrap* (1994) [16]. Nordman restates its geometric-block construction in Section 2.1 of *A note on the stationary bootstrap's variance* (2009) [18]. Following that construction, this implementation restarts at a uniformly chosen observed date or continues circularly from the preceding sampled index.

$$\text{Pr}(\text{restart}) = \ell^{-1}, \quad I_{k+1}^* = 1 + (I_k^* \bmod T) \quad \text{Equation 17}$$

Equation 17 gives the restart probability and the continuation rule when no restart occurs. Here ℓ is the expected block length, T is the phase sample size and the mathematical indices run from 1 to T . On a restart, a new index is uniform on that range. Python uses the equivalent zero-based convention.

Listing 7. Preserving blocks through random restarts and circular continuation

```
indices = np.empty((replicates, observations), dtype=np.int32)
indices[:, 0] = rng.integers(0, observations, size=replicates)
for t in range(1, observations):
    restart = rng.random(replicates) < 1.0 / mean_block_length
    new_start = rng.integers(0, observations, size=replicates)
    indices[:, t] = np.where(restart, new_start, (indices[:, t - 1] + 1) % observations)
```

Source: **market_neutral/inference.py**, function **stationary_indices**. Selected executable lines; surrounding input checks remain in the module.

The primary interval uses 5,000 replicates and expected block length 10; lengths 5 and 20 are retained as sensitivity checks. The reported bounds are the 2.5th and 97.5th percentiles of the bootstrap mean, multiplied by 252. Seeds are specified in the protocol and derive deterministically from phase and block length. No block length is chosen because it yields a preferred conclusion.

8.4. HAC cross-check and interpretation

Newey and West, in *A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix* (1987), develop a covariance estimator using weighted lagged covariances [17]. Their 1986 working-paper version was consulted for the Bartlett weights. Here the intercept-only special case supplies a second estimate of uncertainty for the paired mean.

$$\widehat{\Omega} = \widehat{\gamma}_0 + 2 \sum_{k=1}^K \left(1 - \frac{k}{K+1}\right) \widehat{\gamma}_k \quad \text{Equation 18}$$

Equation 18 uses $K = 10$ lags. Each gamma-hat is the sum of products of mean-centred paired differences k observations apart, divided by the full phase length T . The same T denominator is used at every lag. This convention is checked against the equivalent Bartlett quadratic form.

$$\widehat{SE}(\widehat{\mu}_{\text{ann}}) = 252 \sqrt{\widehat{\Omega}/T} \quad \text{Equation 19}$$

Equation 19 scales the standard error of the sample mean by 252. It does not use square-root-of-252 volatility scaling. The approximate 95% HAC interval adds and subtracts the standard-normal critical value times this standard error.

Listing 8. Accumulating serial covariance with Bartlett weights

```
centered = x - x.mean()
n = len(x)
long_run_variance = float(centered @ centered / n)
for lag in range(1, lags + 1):
    covariance = float(centered[lag:] @ centered[:-lag] / n)
    long_run_variance += 2 * (1 - lag / (lags + 1)) * covariance
```

Source: **market_neutral/inference.py**, function **hac_mean_interval**. Selected executable lines; surrounding input checks remain in the module.

Table 12. Pointwise uncertainty for the primary paired-return difference

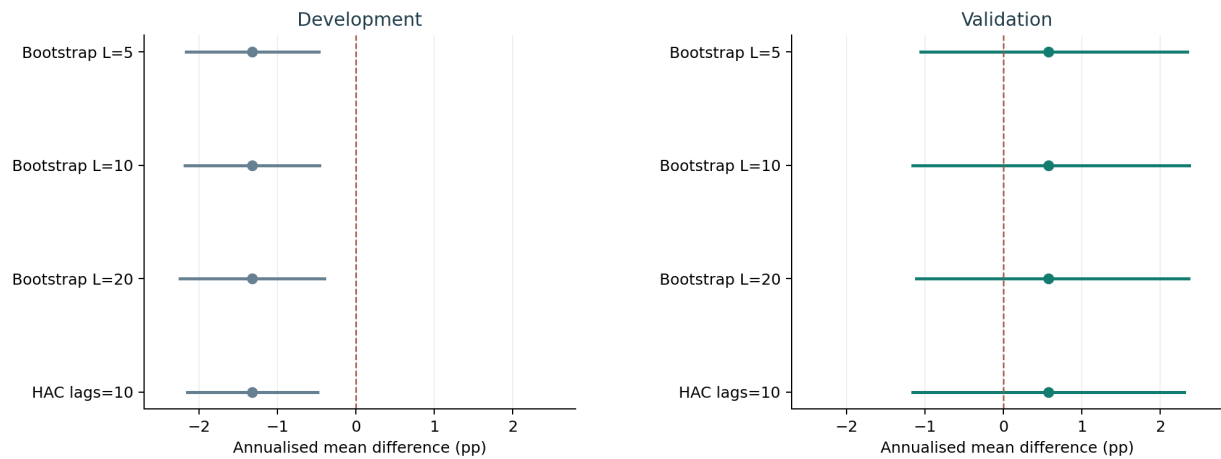
Phase	Mean difference (pp)	95% bootstrap interval (pp)	95% HAC interval (pp)
Development	-1.32	[-2.18, -0.47]	[-2.15, -0.49]
Validation	+0.57	[-1.16, +2.37]	[-1.16, +2.30]

The development intervals lie below zero, while every validation interval spans zero. Thus the modest positive validation point estimate does not establish a reliable graph advantage. These are approximate, pointwise intervals conditional on the observed policies and basket. They assume adequate stationarity and weak dependence, do not refit signals on bootstrapped price paths, and cannot correct survivorship bias, historical selection or repeated model searching.

8.5. Sensitivity and the economic effect of costs

How certain is the graph-minus-baseline difference?

Point estimates and 95% pointwise intervals for 252 times the mean daily net-return difference (percentage points).

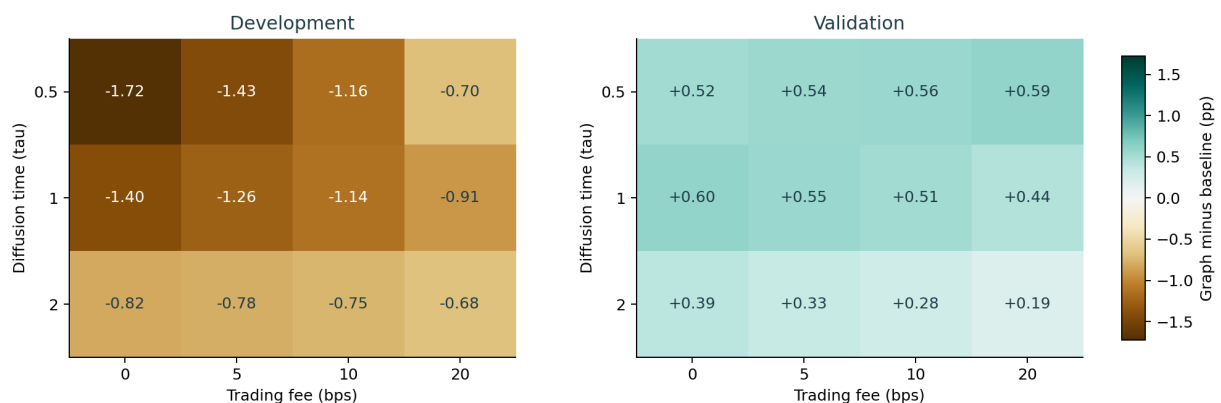


Conditional on the observed policies and basket. These intervals are not CAGR intervals or corrections for strategy selection.

Figure 5. Stationary-bootstrap and HAC intervals for the paired mean

Every declared diffusion-time and fee combination

Cells show graph CAGR minus baseline CAGR in percentage points; 2% annual borrow throughout.



Tau = 1 remains the primary model. These are descriptive sensitivity results, not a parameter-selection exercise.

Figure 6. Graph-minus-baseline CAGR across every declared fee and diffusion time

Every tested diffusion time underperforms the baseline in development and improves its validation CAGR slightly. None is profitable in validation, even when trading fees are zero and borrow remains 2%. This pattern is retained without promoting a different diffusion time to the primary specification. The complete scenario values and calendar-year returns are supplied as machine-readable outputs.

8.6. Gross contribution, borrow and turnover

Separate the signal contribution from recurring costs

Annualised arithmetic contributions at 5 bps and 2% borrow. Gross minus trading and borrow charges equals net.

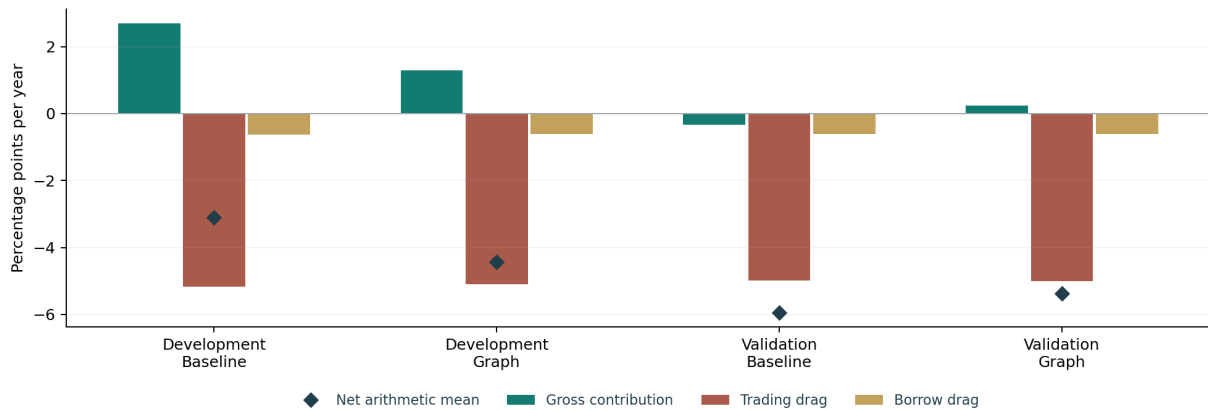


Figure 7. Annualised arithmetic return contributions and recurring charges

For the primary graph in development, the positive gross arithmetic contribution is insufficient to cover trading and borrowing charges. In validation its gross contribution is smaller still. This diagnosis is more informative than reporting a Sharpe statistic alone: the graph changes the relative allocation, but the observed gross signal does not finance the declared turnover cost.

Table 13. Borrow sensitivity: CAGR with trading fees fixed at 5 bps

Annual borrow	Dev. baseline	Dev. graph	Val. baseline	Val. graph
0%	-2.49%	-3.77%	-5.25%	-4.69%
2%	-3.10%	-4.37%	-5.83%	-5.27%
5%	-4.01%	-5.25%	-6.69%	-6.13%

The zero-borrow case retains transaction fees. It is a sensitivity scenario, not an assertion that every short can be financed for free. Likewise, a uniform 5% borrow rate does not represent security-specific availability or a stressed lending book. These tests describe the assumed ledger, not executable brokerage terms.

8.7. Interpreting diffusion through an interactive graph

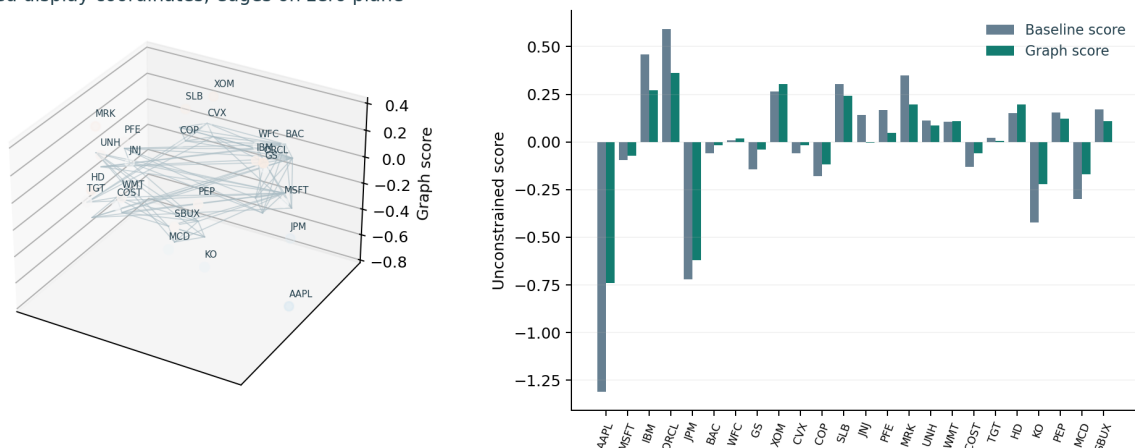
$$g_{\tau}(\lambda) = 1 - \exp(-\tau\lambda)$$

Equation 20

Equation 20 follows directly from Equation 8: an eigenvector of L with eigenvalue λ is multiplied by this factor in x minus its diffused value. The zero-eigenvalue component is removed. Changing τ changes the relative treatment of graph modes; the subsequent exposure projection and scaling remain separate operations.

How diffusion changes the score: 2019-12-31

Fixed display coordinates; edges on zero plane



Display geometry does not enter the strategy. Scores are subsequently projected and scaled; the plotted scores are not trades.

Figure 8. A historical graph and the resulting unconstrained scores

The companion **Graph-Diffusion-Explorer.html** shows 120 monthly graph snapshots, with a rotatable 3D network, date slider and paired score bars. Horizontal node coordinates are a fixed arrangement of the six research groups for display. Lines on the zero plane show retained connections; height shows the graph score. Neither drawing coordinates nor rendered interpolation enter the trading algorithm.

A positive plotted graph score proposes a long direction before the constraints are applied. The final projected weight may have a different sign. This distinction helps connect the mathematical operator to the implementation without presenting raw scores as actual trades or interpreting an attractive network visual as evidence of financial value.

9. Planned persistent-homology extension

Edelsbrunner, Letscher and Zomorodian formalise feature persistence through a filtration in *Topological Persistence and Simplification* (2002) [12]. Their framework motivates measuring how connected components and loops appear and disappear as a scale changes. Persistence is a mathematical description of structure, not by itself a financial prediction.

$$d_{ij,t} = \sqrt{2(1 - \rho_{ij,t})}$$
Equation 21

Equation 21 relates correlation to a distance between standardised return series, following the financial correlation-distance construction associated with Mantegna [6]. It will use a complete valid correlation matrix; the sparse positive-neighbour adjacency matrix must not be substituted for that metric space.

Gidea and Katz, in *Topological data analysis of financial time series: Landscapes of crashes* (2018), study persistence landscapes from sliding-window point clouds of market-index returns [13]. That is relevant precedent, but the planned asset correlation-distance construction here is different. Their findings do not establish that our proposed H0/H1 features will improve trading performance.

The extension will require numerical checks on known shapes, explicit treatment of infinite H0 bars, stable feature definitions and training-only thresholds. Mean correlation and realised volatility will serve as simpler controls. The persistent-homology module remains unimplemented in this milestone.

10. Reproducibility and subsequent milestones

Table 14. Research artifacts and completion state

Artifact	Current status
Historical data and protocol	Cached inputs, source metadata, SHA-256 hashes and guarded period boundaries
Baseline backtest	Reproduced from Milestone 2 within saved rounding tolerance
Graph trading comparison	44 predeclared backtests completed; paired intervals and exposure controls reported
Interactive 3D explorers	120 monthly correlation surfaces and 120 graph/score frames
Persistent homology	Planned; no extracted features or regime results claimed
Final holdout	2020-2025 reserved; not acquired or evaluated
GitHub readiness	Requirements, 25 checks, three notebooks, rebuild scripts and CI configuration
Public repository / LinkedIn page	Planned for final presentation; not published by this milestone

The private research bundle retains the acquired observations for exact offline replay. A public GitHub checkout will exclude provider input caches and regenerate them through the acquisition script. Provider revisions can change a fresh download, so the recorded hashes distinguish exact snapshot reproduction from rerunning the same method on a new data vintage. Permission to redistribute vendor data is not inferred from its technical accessibility.

The paper builder extracts its code excerpts directly from the research modules and generates the contents and lists from captions. Future versions will retain the title page, abstract, abbreviations, symbol definitions, numbered equations, figure and table lists, code listings and author-led references. The final public presentation will explain both the results and the limits of the evidence.

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References 1-14 and 16-18 are journal articles, scholarly books or a peer-reviewed conference paper. Reference 15 identifies the primary data service. The source register records the accessed version and the role of each reference. No citation is intended to imply that another author tested this project's exact strategy.